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Possible constraints on asteroid-mass primordial black holes from Solar system cratering

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ABSTRACT

Due to their novel cratering mechanism, primordial black holes (PBHs) in the asteroid mass window of about 10^{17} to 10^{19} g can produce surface features on Solar system bodies that are distinguishable from traditional impact craters. High-resolution surface scans of old cratered Solar system bodies therefore provide a way to test compact dark matter candidates like PBHs. In this work, we calculate the flux of PBHs across three targets: the Moon, Mercury, and Ganymede. We consider PBHs from both the galactic halo and thick dark disc, and include modulation due to annual and galactic orbits, gravitational focusing, and the vertical oscillation of the Solar system. We show that non-detection of a PBH crater on all three of these bodies has the potential to constrain PBHs as dark matter near the evaporation limit, potentially improving current constraints by up to about an order of magnitude, or roughly 10^{18} g.

Key words: black hole physics – methods: numerical – Moon – planets and satellites: surfaces – Galaxy: kinematics and dynamics – dark matter.

1 INTRODUCTION

Primordial black holes (PBHs), black holes that may have formed in the first instants of the universe, are a popular dark matter candidate (for a review of constraints, see Carr & Kühnel 2022, and for the optimistic 'positivist' perspective, see Carr et al. 2024). Observations now constrain the majority of possible PBH masses to contributing a few percent or less of the dark matter, though a few unconstrained mass windows remain. Of note is the asteroid-mass window, above the evaporation limits of about 10^{17} g and below the microlensing constraints starting at 10^{23} g. This mass window has proven particularly challenging to constrain owing to the weak gravity and small radii; black hole Schwarzschild radii are sub-Angstrom below 10^{20} g.

If the majority of the dark matter is PBHs in the asteroid mass window, then their number abundance would be sufficiently high to expect a few within the Solar system at any given time. With local Galactic dark matter density 8.0×10^{-25} g cm^{-3}, the mean PBH separation in the Galaxy is of order

$$\left(\frac{m_{\rm PBH}}{\rho_{\rm DM}}\right)^{1/3} = 15\,{\rm au}\left(\frac{\rho_{\rm DM}}{8\times10^{-25}\,{\rm g\,cm^{-3}}}\right)^{-1/3}\left(\frac{m_{\rm PBH}}{10^{19}\,{\rm g}}\right)^{1/3}. \quad (1)$$

This fact strongly motivates Solar system searches for low-mass PBH dark matter, and in recent years, the number of Solar system probes proposed in the literature has grown steadily (e.g. Kesden & Hanasoge 2011; Luo et al. 2012; Rahvar 2021; Yalinewich & Caplan 2021; Bellinger et al. 2023; Caplan, Bellinger & Santarelli 2024; Santarelli, Caplan & Bellinger 2024; Tran et al. 2024)

In our recent work, we have performed a detailed calculation of the flux of PBHs through the Solar system and shown that the rate is sufficiently high for a few PBHs to have collided with Solar system bodies such as the Moon (Caplan, Johnston & Santarelli 2023). Such collisions do not destroy compact bodies in the Solar system due to the small Schwarzschild radii and fast encounter velocities, so the survival of planets and moons does not by itself constrain PBH dark matter. Rather, during an encounter the PBH transits the body and accretes matter along a column before exiting with speeds above escape velocity. During the transit the accretion luminosity produces a 'line explosion' resulting in two craters, one on entrance and one on exit, with morphology distinct from traditional rocky impactors and radii of order metres that could be searched for in high-resolution surface scans (Yalinewich & Caplan 2021).

In this work, we further develop our calculations from Caplan et al. (2023) and consider, in addition to the Moon, two new possible targets with old cratered surfaces: Mercury and Ganymede. In Section 2, we present our methods for calculating dark matter fluxes across Solar system bodies from both the halo and a dark disc with complete Solar system and Galactic orbits. We present results in Section 3 and conclude in Section 4.

2 METHODS

To determine the probability of a PBH collision in the Solar system, we must calculate the flux of dark matter through our three targets (the Moon, Mercury, and Ganymede). We consider the kinematics of the dark matter in Section 2.1 and models for the velocity of our targets relative to the Galactic rest frame in Section 2.2.

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2.1 Dark matter flux

Dark matter is assumed to follow a Maxwell–Boltzmann velocity distribution in 3D given by

$$f(\boldsymbol{v}) = \frac{N_{\rm esc}}{(2\pi)^{3/2}\sigma_r\sigma_\phi\sigma_z} e^{-(v_r^2/2\sigma_r^2+v_\phi^2/2\sigma_\phi^2+v_z^2/2\sigma_z^2)}\Theta(v_{\rm esc}-|\boldsymbol{v}|) \tag{2}$$

truncated above the Galactic escape velocity $v_{\rm esc} = 545\,{\rm km\,s^{-1}}$ with the Heaviside function Θ. The normalization $N_{\rm esc}$ is required with the truncation and is given in Caplan et al. (2023); it is of order one percent at most in this work (Binney & Tremaine 2008). For the halo, we use $\sigma_r = \sigma_\phi = \sigma_z = 155\,{\rm km\,s^{-1}}$. Additionally, thick dark discs are a general prediction from cold dark matter models, as early subhaloes merging with the Milky Way at low angle and low velocity get dragged into the disc by dynamical friction. Dark discs are not isotropic, and in this work we use the thick dark disc model from Bruch et al. (2009) and Read et al. (2008, 2009), where $\boldsymbol{\sigma} = (63, 39, 39)\,{\rm km\,s^{-1}}$.

Assuming a constant dark matter density the matter flux from the halo is then obtained by integrating

$$\frac{{\rm d}m_{\rm DM}^{\rm (H)}}{{\rm d}t} = \rho_{\rm DM}^{\rm (H)} \int_{|\boldsymbol{v}|<v_{\rm esc}} \sigma_{\rm eff}(\boldsymbol{v})\,|\boldsymbol{v}|\,f(\boldsymbol{v})\,{\rm d}^3\boldsymbol{v} \tag{3}$$

where $\rho_{\rm DM} = 8\times10^{-25}\,{\rm g\,cm^{-3}}$ is the local halo dark matter density and $\sigma_{\rm eff}(\boldsymbol{v})$ is the effective cross-section of the body due to gravitational focusing, given by

$$\sigma_{\rm eff}(v) = \sigma_{\rm geo}\left(1+\frac{v_{\rm esc}^2}{v^2}\right) \tag{4}$$

with geometric cross-section $\sigma_{\rm geo} = \pi R^2$. The escape velocity is taken to be the Solar system escape velocity at the surface of the body and can be calculated from conservation of energy for the Moon (M), Mercury (Me), and Ganymede (G) respectively as

$$v_{\rm esc}^{\rm (M)} = \sqrt{\frac{2G{\rm M}_\odot}{a_{\rm E}} + \frac{2GM_{\rm E}}{a_{\rm M}} + \frac{2GM_{\rm M}}{R_{\rm M}}} \tag{5a}$$

$$v_{\rm esc}^{\rm (Me)} = \sqrt{\frac{2G{\rm M}_\odot}{a_{\rm Me}} + \frac{2GM_{\rm Me}}{R_{\rm Me}}} \tag{5b}$$

$$v_{\rm esc}^{\rm (G)} = \sqrt{\frac{2G{\rm M}_\odot}{a_{\rm J}} + \frac{2GM_{\rm J}}{a_{\rm G}} + \frac{2GM_{\rm G}}{R_{\rm G}}} \tag{5c}$$

where a is an orbital semimajor axis with additional terms for Earth (E) and Jupiter (J). As an example, the Solar system escape velocity at the surface of the Moon has contributions from three separate potentials: (i) the solar potential at the Earth–Moon system, (ii) the Earth potential at the Moon, and (iii) the self gravity of the Moon. The solar contribution is dominant for all of three bodies. The relative error in $v_{\rm esc}$ when neglecting the other terms is order 10^{-3} for the Moon and Mercury, though it grows to 0.25 for Ganymede when neglecting the Jovian contribution. Escape velocities used are given in Table 1.

Table 1. Orbital velocities, Solar system escape velocities at the surface of the body, and geometric cross-sections used in this work.

	$v_{\rm orbit}$ (km s^{-1})	$v_{\rm esc}$ (km s^{-1})	$R_{\rm geo}$ (km)
Moon	30	42	1737.4
Mercury	47	67.84	2439.7
Ganymede	10.9	21.19	2631.2
Jupiter	13.1	–	–

We now argue that the dark disc can be treated with the same integral as equation (3). The Solar system oscillates in the vertical plane due to the restoring force produced by the mass of the baryonic disc. For a dark disc, the density varies with height z above the centre of the disc; a simple model might be

$$\rho_{\rm DD}(z) = \frac{\Sigma_{\rm DD}}{4h_{\rm DD}}{\rm sech}^2\left(\frac{z}{2h_{\rm DD}}\right) \tag{6}$$

with scale height $h_{\rm DD}$ and surface density $\Sigma_{\rm DD}$ (Buch, Leung & Fan 2019) so that in principle one should use a modified form of equation (3) to account for the vertical oscillation of the Solar system, such as

$$\frac{{\rm d}m_{\rm DM}^{\rm (DD)}}{{\rm d}t} = \int_{|\boldsymbol{v}|<v_{\rm esc}} \rho_{\rm DM}^{\rm (DD)}(\boldsymbol{v},z)\,\sigma_{\rm eff}(\boldsymbol{v})\,|\boldsymbol{v}|\,f(\boldsymbol{v})\,{\rm d}^3\boldsymbol{v}. \tag{7}$$

We remark that the scale height of the thick dark disc is predicted by Read et al. (2008) to be about 1.3 kpc (with the parametrization of equation 6), so over the ≈ 100 pc vertical oscillation of the sun the thick dark disc density should only modulate by at most a few times 10^{-2}. Indeed, the thick dark disc is not thought to be dynamically significant to the thin disc (Read et al. 2008). Uncertainty is therefore dominated not by the scale height and vertical oscillation, but the surface density, as cosmological simulations of galaxy mergers predict the thick dark disc density to be 0.25–1.5 times $\rho_{\rm DM}^{\rm (H)}$ (Read et al. 2008, 2009; Salucci et al. 2010; Vattis & Koushiappas 2018). We therefore take $\rho_{\rm DM}^{\rm (DD)} \approx \rho_{\rm DM}^{\rm (H)}$ as a constant in this work and use equation (3) for the thick disc as well as the halo – this is the dominant source of uncertainty in this work.

2.2 Relative velocity and vertical oscillation

Due to the high directionality of the flux, we must account for both the motion of the Solar system around the Galactic centre as well as the moon or planet's orbital motion relative to the sun. In Caplan et al. (2023), we introduced the following model for the relative velocity of the Moon to the Galaxy in cylindrical coordinates

$$\begin{aligned}\boldsymbol{v} = (&v_\oplus\sin\alpha\cos\beta + v_\oplus\cos\alpha\sin\gamma\sin\beta,\\ &v_\odot - v_\oplus\sin\alpha\sin\beta + v_\oplus\cos\alpha\sin\gamma\cos\beta,\\ &-v_\oplus\cos\alpha\cos\gamma)\end{aligned} \tag{8}$$

where α represents the annual motion of the Moon in the Solar system and β represents the Galactic orbit of the Solar system, both ranging from 0 to 2π over their respective years. The angle of inclination of the Solar system ecliptic with respect to the Galactic plane is $\gamma = 29.8^\circ$, and the galactic orbital speed is taken to be $v_\odot = 220\,{\rm km\,s^{-1}}$ for the halo model (Green 2001) and $v_\odot = 50\,{\rm km\,s^{-1}}$ for the corotating dark disc models (Bruch et al. 2009; Purcell, Bullock & Kaplinghat 2009). As discussed in the derivation in Caplan et al. (2023), this model property treats the Galactic 'seasons' – while the inclination of the ecliptic causes the Solar system to orbit the Galaxy 'face on' at present, our parametrization properly rotates the plane of the Solar system to orbit 'edge on' when β is advanced or retarded by a quarter orbit. Average dark matter fluxes over the lifetime of the Solar system are found by numerically integrating over a grid of orbital phases and averaging.

In our previous work we neglected several corrections to the Galactic orbit including the epicyclic nature of the Sun's orbit in the plane of the Galaxy as well as its vertical oscillations. The epicyclic component of the orbit results in a sinusoidal modulation of the dark matter flux that almost perfectly cancels out when calculating the average flux, and so can be neglected in this calculation. The

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vertical oscillation likewise produces a harmonic modulation that should nearly cancel, but is also straightforward to include as a simple addition to the z component of equation (8),

$$\begin{aligned} \boldsymbol{v}' = (&v_\oplus \sin\alpha\cos\beta + v_\oplus \cos\alpha \sin\gamma \sin\beta, \\ &v_\odot - v_\oplus \sin\alpha \sin\beta + v_\oplus \cos\alpha \sin\gamma \cos\beta, \\ &-v_\oplus \cos\alpha\cos\gamma + v_{\odot,\perp}\cos\delta) \end{aligned} \tag{9}$$

where δ is the phase of the vertical oscillation and is evaluated from 0 to 2π and $v_{\odot,\perp} = 7.8\,\mathrm{km\,s^{-1}}$ is the maximum vertical velocity of the sun (this corresponds to an 80 pc maximum displacement for a sinusoidal oscillation, also see Randall & Reece 2014).

To generalize equation (8) for Mercury we simply replace $v_\oplus$ with its own orbital speed v_{Me}, though the Moon and Ganymede require further treatment. The orbital velocity of the Moon around the Earth is approximately $1\,\mathrm{km\,s^{-1}}$, smaller than the error in the Galactic orbit, and so it was neglected in our previous work. Ganymede requires an additional term as its orbital speed with respect to Jupiter is comparable to Jupiter's orbital speed with respect to the sun, so that equation (9) then becomes

$$\begin{aligned} \boldsymbol{v}' = (&v_{\mathrm{J}} \sin\alpha\cos\beta + v_{\mathrm{J}} \cos\alpha \sin\gamma \sin\beta \\ &+v_{\mathrm{G}} \sin\lambda\cos\beta + v_{\mathrm{G}} \cos\lambda \sin\gamma \sin\beta, \\ &v_\odot - v_{\mathrm{J}} \sin\alpha \sin\beta + v_{\mathrm{J}} \cos\alpha \sin\gamma \cos\beta \\ &-v_{\mathrm{G}} \sin\lambda \sin\beta + v_{\mathrm{G}} \cos\lambda \sin\gamma \cos\beta, \\ &-v_{\mathrm{J}} \cos\alpha\cos\gamma + v_{\odot,\perp}\cos\delta + v_G \cos\lambda\cos\gamma) \end{aligned} \tag{10}$$

where $v_{\mathrm{J}} = 13.1\,\mathrm{km\,s^{-1}}$ and $v_{\mathrm{G}} = 10.9\,\mathrm{km\,s^{-1}}$ are the orbital velocities of Jupiter and Ganymede respectively, and λ parametrizes the phase of Jupiter's annual motion, again evaluated from 0 to 2π. We list the orbital velocities, escape velocities, and geometric cross-sections that we use in Table 1.

3 RESULTS

We now calculate the average dark matter flux through the Moon, Mercury, and Ganymede.

3.1 Halo

We integrate equation (3) with equations (9) and (10) using 10 values evenly spaced from 0 to 2π for all of α, β, and δ, a total of 1000 integrations, as these sample the modulation from the orbits and the sinusoidal oscillation in the case of δ. For the Moon and Ganymede, our grid of integrations is expanded to include λ. We compute the average dark matter flux from this grid of integrations.

Average dark matter fluxes are given in Table 2. When comparing our results from halo models for the Moon to those in Caplan et al. (2023), there is no change when vertical oscillations are included. This is due to the largely sinusoidal motion of these oscillations through constant dark matter density, resulting in no net flux difference when averaged. Meanwhile, we find that Mercury and Ganymede experience approximately a factor of two greater flux than the Moon. This difference is dominated by the difference in geometric cross-section. Indeed, ratios of fluxes are comparable to ratios of the geometric cross-sections of the bodies.

To understand how gravitational focusing enhances the fluxes, we also repeat our calculation with gravitational focusing disabled, so that $\sigma_{\mathrm{eff}} = \pi R_{\mathrm{geo}}^2$. Results are again reported in Table 2, and show that gravitational focusing is responsible for a modest enhancement of almost 10^{-1} in the flux through Mercury from the halo as it has the largest v_{esc}, while the smaller v_{esc} of the Moon and Ganymede result in weaker enhancements of order 10^{-2} and 10^{-3} respectively.

Table 2. Average dark matter mass fluxes (in g s^{-1}) for each body using each disc model with and without enhancements from solar focusing. Uncertainty in halo fluxes is of order 10 per cent, while uncertainty is dark disc fluxes is of order 50 per cent from uncertainty in the dark matter density.

	Halo	Thick dark disc
Moon	2.277	0.895
Moon (no focusing)	2.227	0.718
Mercury	4.649	2.361
Mercury (no focusing)	4.394	1.503
Ganymede	5.135	1.705
Ganymede (no focusing)	5.106	1.599

3.2 Dark discs

The fluxes for the thick dark disc are given in Table 2. For the dark disc, we perform our calculations in the same manner as with the halo, except we now use solar velocity relative to dark disc $v_\odot = 50\,\mathrm{km\,s^{-1}}$ and velocity distribution $\sigma = (63, 39, 39)\,\mathrm{km\,s^{-1}}$ mentioned in Section 2.1, as well as the same uniform dark matter density as with the halo. We emphasize again that factor of a few uncertainties remain in the density of a thick dark disc (Read et al. 2008, 2009). Like with the halo, adding the vertical oscillation to our model has no effect on the flux through the Moon for the thick dark disc model. Gravitational focusing has a significantly greater effect on the dark disc as the relative velocity to the sun and velocity dispersion are comparable to the escape velocities. In the absence of focusing, the flux from the dark disc is approximately a third that from the halo. Focusing yields an order 10^{-1} enhancement, with the greatest effect seen with Mercury because the effect scales with v_{esc}.

3.3 Poisson probability of PBH collisions

PBH collisions are a discrete Poisson process and have a probability of at least one collision in time τ given by $P(N \geq 1) = 1 - e^{-R\tau}$ with rate constant R. We take $\tau = 1.44 \times 10^{17}$ s to be the age of the Solar system. We define R as the average flux from Section 2.2 divided by the PBH mass such that $R = (f\,\frac{\mathrm{d}m_{\mathrm{DM}}}{\mathrm{d}t})/(m_{\mathrm{PBH}})$, with f the PBH fraction of dark matter. In this sense, the small mean dark matter flux calculated in the previous sections becomes a proxy for the probability flux for a PBH collision. As we are interested in further constraining PBH masses, it is then trivial to calculate the maximum PBH mass such that at least one collision has occurred in the history of the Solar system with 95 per cent probability,

$$m_{\mathrm{PBH}}\,(P(N \geq 1) \geq 0.95) = \frac{\tau f}{3}\,\frac{\mathrm{d}m_{\mathrm{DM}}}{\mathrm{d}t} \tag{11}$$

using $\ln(0.05) \approx -3$. In Table 3, we show the largest PBH masses that can be constrained at 95 per cent confidence if $f = 1$. If no dark discs exist, then the Moon alone can constrain PBH dark matter up to about 10^{17} g and individually Mercury and Ganymede could constrain up to about 2 to 3×10^{17} g. When including a thick dark disc, the roughly 50 per cent increase in dark matter flux translates to a factor of 50 per cent greater PBH masses that can be constrained. As an upper limit, combined observations of the Moon, Mercury, or Ganymede could probe PBH dark matter up to about 10^{18} g.

We present the existing constraints from evaporation, microlensing, gravitational waves, and accretion on Fig. 1. We overlay the possible constraints from this work for each of the three target bodies, in addition to a constraint for non-detection on all three. Note that

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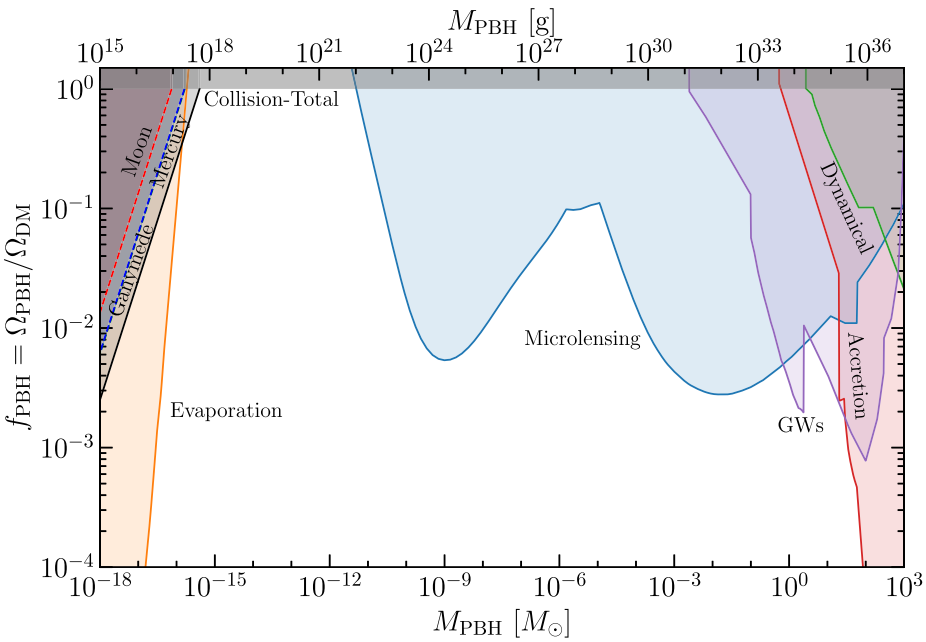

Figure 1. Existing constraints on PBHs as a fraction of total dark matter $f_{\rm PBH}$, along with our theoretical constraints in the event of PBH crater non-detection on the Moon (leftmost, dashed), Mercury (center, dashed), Ganymede (center, dashed), and a combined constraint from all three (rightmost, solid). Note that the line indicating Mercury's constraints is very close to that of Ganymede and is thus largely overlapped by it.

Table 3. Maximum PBH mass constraints that can be obtained by non-detection of a PBH crater assuming $f = 1$ at 95 per cent confidence.

	Halo only (g)	Halo and thick dark disc (g)
Moon	1.1×10^{17}	1.5×10^{17}
Mercury	2.2×10^{17}	3.3×10^{17}
Ganymede	2.4×10^{17}	3.2×10^{17}

these constraint curves use the combined flux from the halo and the thick dark disc.

3.4 Crater radii

We now consider the characteristic sizes of craters produced by PBHs. Dissipative forces cause the transiting PBH to deposit energy in the surrounding medium producing a line explosion. We consider dynamical friction and accretion as possible sources.

The energy deposition from dynamical friction of a PBH passing through the earth is given in equation (21) of Rahvar (2021) and provides a useful starting point for any rocky body. For a fiducial 10^{18} g PBH from the halo, the energy deposition per length is approximated by[1]

$$\mathrm{d}E/\mathrm{d}x = 5.8 \times 10^4 \mathrm{J\,m^{-1}} \left(\frac{m_{\rm PBH}}{10^{18}\,\mathrm{g}}\right)^2 \left(\frac{v_{\rm PBH}}{220\,\mathrm{km\,s^{-1}}}\right)^{-2}. \quad (12)$$

This is small, but for a PBH from the dark disc ($v_{\rm PBH} \approx 50\,\mathrm{km\,s^{-1}}$), this grows to 1.9×10^6 J m^{-1} – within 1 m of the surface this is approximately 0.5 kg TNT equivalent. One might expect the scale of the resulting crater to be comparable to a buried hand grenade or small land mine. Therefore, dynamical friction would not be expected to produce a detectably large crater on its own.

We now consider accretion onto the PBH. The accretion efficiency is highly uncertain, but simple schemes for estimating the accretion rate and luminosity are given in Caplan et al. (2023, 2024). A PBH moving at $v_{\rm PBH}$ through a static medium can accrete at a maximum Bondi-like rate that we approximate by $\dot{M}_{\rm B} \approx 4\pi(GM_{\rm PBH})^2\rho/(c_s^2 + v_{\rm PBH}^2)^{3/2}$ with medium density $\rho \approx 2.5\,\mathrm{g\,cm^{-3}}$ and sound speed $c_s \approx 5\,\mathrm{km\,s^{-1}}$, respectively. Since $c_s \ll v_{\rm PBH}$, the medium sound speed drops out and we estimate the maximum possible accretion rate to be

$$\dot{M}_{\rm B} \approx 13\,\mathrm{g\,s^{-1}} \left(\frac{m_{\rm PBH}}{10^{18}\,\mathrm{g}}\right)^2 \left(\frac{v_{\rm PBH}}{220\,\mathrm{km\,s^{-1}}}\right)^{-3} \left(\frac{\rho}{2.5\,\mathrm{g\,cm^{-3}}}\right) \quad (13)$$

or $1100\,\mathrm{g\,s^{-1}}$ for a slower PBH from a dark disc.

Now, consider the Eddington luminosity, $L_{\rm Edd} = 4\pi G m_{\rm PBH} c/\kappa$ with opacity κ. If the opacity of the accreted silicates is perhaps an order of magnitude greater than ionized hydrogen then the accretion luminosity is perhaps $10^{15}\,\mathrm{J\,s^{-1}}$. From the definition of the accretion luminosity,

$$L = \frac{\epsilon}{1-\epsilon}\dot{M}_{\rm BH}c^2 \quad (14)$$

assuming accretion efficiency $\epsilon = 0.1$, the mass accretion rate needed to maintain the Eddington luminosity is only $60\,\mathrm{g\,s^{-1}}$.

Taken together, this analysis suggests that a 10^{18} g PBH from the halo transiting a rocky body in the Solar system may be accreting at or near the Eddington limit, and is limited to consuming the matter content of the column swept out by the Bondi radius which is at most a few micrometres. The accretion energy is deposited effectively instantaneously in this micrometre-width column given that the crossing time is much faster than the sound speed, and this is true whether the energy is generated from radiatively limited accretion (Eddington) or sound speed limited accretion (Bondi). Following

[1]Rahvar (2021) primarily works with the transit time-scales and therefore it was convenient in that work to report the total energy of the transit and assume the transit spans an earth radius. The expression below is therefore obtained by dividing out the earth radius from equation (21) of Rahvar (2021).

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this, the line explosion proceeds and shocks the surrounding metres of material. Note that our previous work did not consider Eddington accretion, though including it does not revise the predictions in Yalinewich & Caplan (2021) or Caplan et al. (2023). We considered only the Moon in those works, and the smaller target size restricted our focus to 10^{17} g PBHs, which are not massive enough to cross the transition from a Bondi-like accretion ($L \propto M_{\rm PBH}^2$) to an Eddington-like accretion ($L \propto M_{\rm PBH}$). Estimating the energy deposition per length, $L/v_{\rm PBH} \approx 4.5 \times 10^9\,{\rm J\,m^{-1}}$, roughly a ton of TNT equivalent energy per metre.

Solutions for cylindrical explosions and shocks are abundant in the literature, and one might expect a paraboloidal shock envelope to follow the PBH as the shock expands outward and slows (Taylor 1950; Sakurai 1953; Lin 1954; Sedov 2018). However, cylindrical explosions in a medium at a boundary with vacuum do not have readily available solutions and have only briefly been studied numerically with smoothed particle hydrodynamics (Yalinewich & Caplan 2021). We instead estimate crater radii by generalizing from empirical laws for point explosions, as crater dimensions as a function of explosive yield and depth are well studied for nuclear weapons and other conventional munitions (Glasstone & Dolan 1977). For a point explosion, the crater dimensions can be estimated from the scaled depth of burst, proportional to $x/\bar{E}^{1/3}$ where x is the burst depth and $\bar{E}$ is a dimensionless scaling of the energy (typically $\bar{E} = (E/E_0)$ with E the weapon yield and $E_0 = 1$ kiloton TNT equivalent). In principle, one could imagine integrating many stacked point explosions of low yield to obtain a point explosion of equivalent scaled depth and yield. However, given the large uncertainties, it is perhaps more straightforward to observe that within a few metres of the surface the deposited energy is comparable to the yield of a few tons of TNT, and by the known point explosion scaling that should excavate a multiple of a few times the TNT equivalent mass. Craters produced by the surface detonations of a few tonnes TNT studied by Xu et al. (2021) have basin radii of order metres and comparable heights (difference between maximum basin depth and maximum rim height above the original ground). We argue that crater basin radii up to 10 m may be typical for the PBHs considered in this work.

This radius is only a rough estimate. We emphasize that the nature of the line explosion will necessarily result in a different basin profile that depends strongly on the fallback and jamming of the ensuing granular flows, and can be challenging to distinguish from craters of other high-energy sources if looked at alone. Beyond the basin, we expect the unique ejecta blanket profile found by Yalinewich & Caplan (2021). The uncertainty in the energy deposition is large, and varies with parameters including the exact substrate density (and especially the granularity and grain packing near the surface), the poorly known accretion efficiency, and multidimensional flows in the accreted matter. Early craters will be further shaped by later impact-induced seismic activity and solar wind sputtering. This last factor may pose the greatest challenge to identifying craters from line explosions. For example, Speyerer et al. (2016) argue that the top 2 cm of lunar regolith is overturned with a characteristic timescale of 81 000 yr and Fassett et al. (2017) find that crater degradation may be twice as fast on Mercury as it is on the Moon. Detailed topographic diffusion models are needed to understand how a PBH crater evolves, and if it remains distinguishable from traditional point explosion impact craters.

4 DISCUSSION

The presence or absence of PBH craters on old cratered Solar system bodies provides a test of PBH dark matter using extremely large planetary-scale detectors whose integrated detection times rival the solar age. In this work, we have improved upon our previous calculations of the dark matter flux through Solar system bodies and show that it is likely that a PBH up to 10^{18} g has hit a rocky body in the Solar system if PBHs of this mass comprise the majority of the dark matter. While there is significant overlap between our Solar system constraint in Fig. 1 and the evaporation constraint, we emphasize that the new constraints proposed here are completely independent of Hawking radiation which remains hypothetical. We further emphasize that the craters produced by PBH transits are expected to be small, order metres, but near the resolution limit of current generation satellites like the Lunar Reconnaissance Orbiter. However, we emphasize that non-detection from one of these hypothetical searches does not necessarily imply non-collision, given that the surface features we are searching for no doubt evolve on million or billion year time-scales. That being said, the possible constraint on PBHs described in this work would be a 'free' result if these scans do become available as a result of future space missions. On a personal note, it is the opinion of the authors that a non-detection may not prove to be a strong constraint, but identifying any candidates would be compelling even though the prospects are poor. A positive detection simply requires a pair of craters that could be verified with a sample return mission to search for a shocked column of matter connecting them, and if found would be unambiguous evidence for compact dark matter.

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DATA AVAILABILITY

No new data were generated or analysed in support of this research.

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